排除 84 元配置，将上界从 84 改进到 83

完整结构归约、实际资源证明与精确有限核验

Sum-of-Subset 研究项目

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摘要

从提交 `cf26c50` 已独立闭合的上界 84 出发，排除大小恰为 84 的可行集合，得到 $71 \le N \le 83$。本次不再只给出条件标量系统：重新计算大小 84 的共同整数源归约， 处理等差支撑的边界，给出全部高洞情形的区间延拓；低正洞统一局部化，随后按空覆盖单元数处理。 主要新步骤是两空单元的统一证明、四空无所有者族的两次截断，以及五空 alpha 族 先证明次序、后证明正常性的扣减链 $14 \to 27 \to 31 \to 35 \to 38 \to 42$。 五空正洞只需微小剩余预算。报告明确区分无限集合论证、有限参数验证和组件分类。 验证器使用 Python 标准库及整数或有理数运算；它不是形式化证明助手。

Contents

1 结论、前提与证明接口 **3**
1.1 记号与实际覆盖 3

2 共同格子与结构归约 **3**
2.1 不需要本原性的低容量例外引理 4

3 源包络、实际截断与区间延拓 **5**
3.1 所有高洞 $z \ge 15$ 5
3.2 低正洞的统一局部化 6

4 从活动判据到完整终端 **6**
4.1 一空、两空及六空 6
4.2 三空与四空的实际接入 7
4.3 五空的全部分支 7

5 覆盖、验证边界与复现 **8**

A Size84 structural reduction and all high-hole strata **9**
A.1 Fresh parameters and layer identities 9
A.2 Small boundaries 9
A.3 AP branches and low-c lemma 9
A.4 Both nonAP inverse repairs 10
A.5 Complete high-hole stratum 10

B Low-hole localization and the one-hole terminal **12**
B.1 A simpler localization, valid for every j >= 1 12
B.2 P1: reuse the sharp one-hole terminal 12
B.3 P2 12

C A shared two-hole proof: Z2 and P2 at size84 13
C.1 Restore leading activity in Z2 . . . 13
C.2 P2 activity and a minimal positive-core defect . . . 13
C.3 No two caps . . . 14
C.4 Four exact beta/O exclusions . . . 14
C.5 Exhaustive remaining holes and a common terminal . . . 14

D Positive-hole activity and terminal proofs 16
D.1 One dense seed and a ternary prefix census . . . 16
D.2 Activity interval (j >= 3) . . . 16
D.3 Fresh two-cap identity and j3/j4 grammar . . . 16
D.4 j3 terminal bridge . . . 17
D.5 j4 terminal bridge . . . 17
D.6 j5 complete terminal . . . 18

E Fresh geometric derivations for Z3/Z4 terminal cards 19
E.1 Z3 A0 . . . 19
E.2 Z3 A1 . . . 19
E.3 Z4 physical capacities and masters . . . 20
E.4 Z4 O: deriving the previously conditional constant994 . . . 20

F Size84 Z4 ownerless terminal 22
F.1 Actual fixed credit and first H gaps . . . 22
F.2 Both H have core hits, both alpha clips light . . . 22
F.3 Other H/alpha statuses . . . 23

G Complete local Z5 reduction at size84 (common source assumed) 24
G.1 The base holes before outer integrality . . . 24
G.2 Integral coordinates and the literal grammar . . . 24
G.3 Common endpoint and forest calculus . . . 25
G.4 A0 . . . 25
G.5 A1/B transports reconstructed at d76 . . . 26

H Size-84 Z5 alpha: order before normality 28
H.1 The initial tail credit does not need normality . . . 28
H.2 Separate the two order tests . . . 28
H.3 Normality and terminal . . . 29

1　结论、前提与证明接口

定理 1.1. *若有限实数集 A 满足 $\mathrm{FS}(A)\subseteq A+A+A+A$，则 $|A|\le 83$。结合仓库已验证的 71 元构造，有* $\boxed{71\le N\le 83}$。

子集和包括空集和，四项和允许重复。已有上界 84 作为归纳前提；若 84 元反例不含零，加入零会产生 85 元反例。因此可假设零属于反例。为每个子集和选定一个四项表示，这些表示构成有限个齐次整数线性方程。其有理核在实核中稠密，连同指定坐标为零的方程，可取保持所有严格次序及符号的有理解，再乘正公分母。因此只需排除一个含零的 84 元整数反例；并未断言每个原实数集本身有公度。

本报告沿用已经逐项核查的通用引理：四极值覆盖与剥离、非等差缺陷递推、计数洞引理、有效洞引理、同时逆定理及短直径定理、整数四极值限制和集界、实际资源分组损失引理。依赖源文本包括旧完整报告的附录 D/E、`STRUCTURAL86.md` 所列通用定理，以及大小 85 的结构伴随证明；本次仅复用其通用论证，所有与基数有关的有限参数重新计算。完整路径及 SHA-256 见 `data/upper83_complete_sources.json`。旧上界 84 的 140 项有限义务由原完整入口重新运行；不能以旧 85 元分类表代替新 84 元分类。

1.1　记号与实际覆盖

去掉两侧各四个外点，得 $|D|=76$；再做两次两侧各两点的剥离，得

$$E=P_s\cup\{0\}\cup(-P_t),\quad |E|=68,\quad r_s+r_t=67.$$

标量分析可反射为 $r_s=33-k, r_t=34+k$，但所有者证明始终保留物理符号。令 $T_r=r(r+1)/2$，$I=k(k+1)=(r_v-33)(r_v-34)$。核心点写成 $x_i=i+d_i$，$d_i\ge 0$ 不减；定义

$$\tau_v=\sum d_i,\quad \xi_v=d_{r_v},\quad R_v=r_v+\xi_v,\quad R=67+\xi,\quad 0\le\xi\le\tau.$$

每侧四个源尾点为 $R_v{+}a_v, R_v{+}b_v, R_v{+}c_v, R_v{+}e_v$，偏移为严格递增正整数。令 $M_v=R_v{+}e_v$，$L=e_s+e_t$，$M=R+L$，$U=\sum_v(a_v+b_v)$，$C=c_s+c_t$，$J=L-C$。S_v 是 D_v 正点总和，其中 $D_s=D, D_t=-D$。外点为 $v_1>v_2>v_3>v_4>M_v$，定义

$$d_v=v_1-v_2,\ g_v=v_3-v_4,\ h_v=v_2-v_3,\ X_v=v_4-M_v,\quad t_v=d_v-M,$$

$$\lambda_v=d_v-v_3-v_4,\quad U_v=d_v-v_4,\ V_v=d_v-v_3,\ H_v=d_v+h_v-v_4,\ O_v=H_v+d_v,$$

$$\alpha_v=h_v-v_4,\quad \beta_v=\alpha_v-d_v,\quad C_{iv}=\lambda_v-o_{i-3}\ (4\le i\le 7).$$

这里普通中心 U_v 与源偏移和 U 不同，固定标签 U_i^v 又是另一对象。

两个首项资源为 $\lambda_v+(D_v+D_v)$，容量 $K=|D+D|$。普通覆盖有 18 个整单元，每个容量 76：C_4 始终配对；分支 A 在 $\beta_s+\beta_t\le 0$ 时配对 beta，分支 B 配对 C_5。一个配对只拥有一次容量，只有实际唯一贡献侧才能按原生普通资源计损失。固定 110 个标签中的 26 个互不相交负和组给出固定容量 84。若有 j 个空整单元，令 $c=76j+\delta$，则

$$S_s+S_t=2K+(18-j)76+84+w-\delta,\qquad \delta\ge 0. \tag{1}$$

w 是 $\mathrm{FS}(D)$ 的整数凸包洞数；两物理目标的洞数之和是 w。对互不相交的实际资源组，其容量减去目标并集的损失之和不超过 δ。相容额外固定扣减 e 将预算降为 $\varepsilon=\delta-e$。

2　共同格子与结构归约

定义

$$\mu_X=|\,\mathrm{FS}(X)|-1-T_{|X|},\quad \rho_X=\mu_X+2-2\Delta_0(X),$$

其中 $\Delta_0(X)$ 数 X 中不能在 $X\cup\{0\}$ 中表示为不同元素和的对角点。令 $\kappa=|\mathsf{FS}(P_s)+\mathsf{FS}(P_t)|-|\mathsf{FS}(P_s)|-|\mathsf{FS}(P_t)|+1$。通用剥离恒等式在本基数成为

$$\Lambda=1157+I,\quad \Omega=612-I,\quad \rho+\kappa+2\theta+c+\ell_1+\ell_2+g=\Omega.$$

这里所有松弛非负，ρ 在小边界允许已知下界 -8。两侧符号不能消失：初始单侧下界 $1+T_{68}=2347>2093$，后续下界 $1+T_{67}=2279>1901$。未配对点满足 $q^2\le 2977$，奇数 $q\le 53$，故至少七对相反点；$\Omega\ge -8$ 进一步给较小侧至少九点。

获得共同核心格子以后、尚未使用整数四极值不等式之前，

$$|\mathsf{FS}(D_1)|-2|\mathsf{FS}(E)|\le -125-2I+2z,$$

小侧不超过 14 时右侧加 16。负上界同时强制下一层同格和两极端平移相交。外层由 $\Lambda>294+\Omega-\rho$ 强制同样结论。完成这两步之后才能使用 $G=2\cdot 76-10=142$，得到

$$\mathcal{T}\ge \mathsf{max}(48,z,142-2z),\quad \mathcal{T}\ge \kappa+142-3z,\quad c+\rho+\mathcal{T}\le 612-I,\quad w\le 4z. \tag{2}$$

$h_s,h_t\le\kappa$，$z=h_s+h_t-\kappa\le \mathsf{min}(h_s,h_t)$。以下是完整归约的关键修复，详细参数和边界在附录 A。

1. 较大支撑为 AP、较小支撑非 AP 时，使用 $\rho_s\ge n(r_s)$ 后得到 $\kappa<T_{r_t}$，强制单一剩余类和 $z=0$。两支撑都为 AP 时，较小侧 $a\le 32$ 仍有 $\Omega-\rho_s<T_b$；仅 $a=33,b=34$ 需另论。此时互素内禀步长只有 $(1,1),(1,2),(2,1)$，不等步长导致 $z=0$、$\kappa\ge T_{33}$，由后述小 c 引理排除。不能在所有不平衡两 AP 行直接声称 $\Omega<2T_a$。

2. 仅较小支撑 AP 时，正确阈值是 $\kappa\le\Omega-\rho_s-(b-5)<2T_a$。单一剩余类的截断长洞或唯一离格生成元代价超过预算。两剩余类是二阶剩余类；规范为偶数 AP 后，最小奇生成元 W 的质量下界

   $$\mu_t+\kappa\ge T_a+1+(b-t)(b-t+1)+tW+t(t-1)-W-T_b$$

   在 $W=19$ 已超过允许量，且对 W 斜率 $t-1>0$。故 $W\le 17,z\le 16$，格化后 $c\le 93<152,w\le 64$。非本原单侧 AP 的相应界为 $c\le 85<152$。

3. 两支撑非 AP 时，较大侧严格逆条件自动成立；较小侧充分条件仅 $k=0,1$ 自动成立。实际不成立的 $k=2,\dots,21$ 使用新计算的 20 个计数端点修复。每行核对起点、离格、后继区间拼接和短直径的严格余量；主范围修复后 $z\le 18$。较小非本原侧给 $c\le 65<76,w\le 72$，由全非空引理排除。

4. 实际严格逆条件成立时，$z\le 62+I$ 已使内层比较严格；其余 $z\ge 63+I$ 使用与基数无关的 $h\ge 49$ 轮廓。若一侧最后两平移不交，其代价 $T_{r-2}+2-2r$ 加另一侧轮廓与 $63+I$ 超过 Ω，38 个定向行最小余量为 77。于是两侧末次平移都相交，最大元删除将内层比较加强为 $-125-2I<0$。这里未提前使用 $G=142$。

$a=9,\dots,14$ 的 AP、实际严格逆和端点修复边界全部由附录 A 分别排除。最终获得两侧本原的共同整数源，$15\le r_s,r_t\le 52$。若 $z>0$，两侧支撑均非 AP，故

$$\mu\ge 61,\quad \tau=\mu+h_s+h_t\ge 61+2z. \tag{3}$$

2.1 不需要本原性的低容量例外引理

通用守护空隙计数在全非空时给 $w\ge 2M+10\ge 160$。对上述 $c<152,w\le 64$ 的结构例外，若 $j=0$ 即矛盾；若 $j=1$，$\delta\le 75$。一个空整单元的原生中心恒等式确定整性，实际 C_6 或 C_7 给 $d_v\ge 80$。原始帽形界 $t_v\le 12+\eta_v\le 76<80$、$2t_v\le 17+\eta_v\le 81$ 足以得到唯一所有者与六边森林。源界给 $\tau\le 444$；全秩核中若 $\phi_s+\phi_t\ge 42$，质量至少 $2T_{21}=462$，故 $F\ge 27$。于是 $2X+G\ge 6F-\delta\ge 87$，但所有者恒等式给 $2X+G\le (17+64)/2-2=38.5$。此处只用 $\eta_v\le w$，未提前假定对称洞数。

3　源包络、实际截断与区间延拓

令 $A_{*}=I+\tau+\delta$。通用源定理给

$$\Theta=76j-296-w+A_{*}\le 160,\qquad \Theta+B_{s}+B_{t}\le 170,$$

其中 B_{v} 是四个源偏移两两和超过最大偏移的实际不同值数。 结合删除一或两个最大源点的包络，得到

$$A_{*}\le 456-76j+w,\qquad L\ge \Theta-8, \tag{4}$$

$$J+U\le 328-\Theta,\qquad J\ge \Theta-18, \tag{5}$$

$$K\ge \Theta+2R+2,\qquad K-M\ge R+76j-454-w+A_{*}, \tag{6}$$

$$\max(4,\Theta-88)\le e_{v}\le 325-\Theta. \tag{7}$$

$j=5,w=0,\delta\ge 10$ 时另有尖锐加强 $L\ge A_{*}+78$。 这些源结论在先前全秩报告中已有完整证明；本次用相同实际定义。

对每一侧选择某个真实前缀，令

$$\phi_{v}=\min_{j}(j+d_{r_{v}-j}),\qquad F=68-\phi_{s}-\phi_{t}.$$

所得同一个实际子核心 Q 在整数位移 x 下满足 $|Q\cap(Q+x)|\ge F-x$。按所需的 $\beta F+q_{0}|Q|$ 选择一个优化前缀， 而不能独立优化 F 与 $|Q|$。逆阶梯质量公式给

$$\tau+F\ge 68,\ \tau+2F\ge 134,\ \tau+3F\ge 198,\quad \tau+4F\ge 260,\quad \tau+6F\ge 378,$$

$$2\tau-2\xi+7F\ge 444,\qquad 2\tau+6F\ge 396.$$

森林按实际目标并集计损失；若一个分离臂中有空顶点，其容量为零。 所有未使用的原生损失返还预算，绝不向空配对指定所有者。

3.1　所有高洞 $z\ge 15$

令 $H=h_{s}+h_{t}=\kappa+z=\tau-\mu$，核心中央区间长度参数 $W_{0}=1156+I+\mu-H$。只要两侧核心质量都大于 H， 简单和集洞引理给中央区间 $[H,C_{0}-H]$，$C_{0}=1156+I+\tau$。 新的外层恒等式和两最大元源包络给

$$M\le 836+H+(\rho-\mu)/2+z-I-c,$$

$$W_{0}-M\ge -904+4I+3c+\tfrac{3}{2}(\mu+\rho)-3z. \tag{8}$$

完全重放的通用稠密轮廓提供以下成对下界；H 上界还应减 $I+c$。

z	μ	ρ	$I+c$ 上界	H 上界	$W_{0}-M$ 下界
15–16	418	288	214	246	107
17–20	461	331	179	219	224
21–48	503	373	191	289	266

由 $\min_{r}(T_{r}+I)=385$，三段各侧核心质量超过 H 的余量至少为139、166、96。 在 $z\ge 49$，$\mu\ge 539,\rho\ge 409$，$\kappa\le 203-I-c$、 $H\le 406-2I-2c$，各侧质量余量至少56。 若 $j\ge 1$，$c\ge 76$，宽度余量至少 $365+7I+6(c-76)>0$。 若 $j=0$，原始 $w\ge 2M+10$ 与 $w\le 4z$ 给 $M\le 401$，而 $W_{0}\ge 1289$。 因此八个源尾点逐个加入时，中央区间均能拼接；共同边界洞定理给偶数 $w\le z$。 $j\ge 1$ 时 $A_{*}\le 380+w$ 与 $\tau\ge 418+2z$ 矛盾； $j=0,z\le 48$ 与 $w\ge 160$ 矛盾，尾段则由 $\tau\ge 539+2z>456+w$ 排除。

3.2 低正洞的统一局部化

对 $1 \le z \le 14, j \ge 1$，先只用 $w \le 4z \le 56$。 $\min(T_r + I) = 385$、$\max(r - I) = 34$ 和 $\mu \ge 61$ 给

$$C_v - H \ge 76j - w - 10 + \delta > 0,$$

$$W_0 \ge 822 + 76j - w + 2I + \delta \ge 842, \quad M_v \le 655 - 76j + w - I - \delta \le 635.$$

故同样逐个延拓八个源尾点，得到偶数 $w \le z \le 14$、$\eta_s = \eta_t = w/2$， 所有目标洞在末端边界条带。$j = 0$ 和 $j \ge 6$ 已分别由全非空界和正质量下界排除。

4 从活动判据到完整终端

零洞用 Z_j，低正洞用 P_j 表示。 守护空隙的三个实际帽形界为

$$O_v \text{ 活动} : t_v \le 12 + \eta_v, \quad \beta_v, H_v \text{ 活动} : t_v \le 9 + \eta_v, \quad \beta_v, O_v \text{ 活动} : 2t_v \le 17 + \eta_v.$$

它们只在已证明的正首项最大值或相应普通中心守护条件下使用。 完整目标的整数活动区间为 $[1 - M_v, S_v]$；正洞的相同区间由边界条带、 至少 18 个相反核心像与至多七个目标洞证明。 附录 B–H 逐一给出整性、活动链、资源名单与终端，以下突出新步骤。

4.1 一空、两空及六空

Z_1, P_1 的实际所有者语法留下 beta、H、O 同侧活动。 带首项根损失的六边森林给

$$2t_u \ge I + \max(99 - 5\xi + \tau, 27 + 2\xi) - w \ge 51 + I - w,$$

与 $2t_u \le 17 + w/2$ 矛盾，低正洞余量至少13。

Z_2 先补足可能空首项的情形：若一首项空，$\delta \ge K$ 与 $K - \delta \ge I + \tau - 8$ 强制 $I + \tau \le 8$。 剩余普通损失不超过8，整性恒等式使它们正常。 空首项若 HIGH 至少两个普通资源各损失 $r_v + 5 > 8$；若 LOW， 跨侧链强制分支 B 恰好空 C_{6v}, C_{7v}，正常 V_v 给 $X_v \ge 68$， 相反活动 C_7 与 O 的帽形界矛盾。因此两首项都活动。

P_2 另有 $\tau - \xi \ge 32$：第一个绝对核心子集和洞小于最大核心点 ≤ 352。 至少34点的一侧删去最大点后质量至少 $T_{33} = 561$，其前缀仍缺这个洞。 任意 n 个不同正整数若子集和不是完整区间，其质量超出 T_n 至少 $n - 1$。 用首次贪心失败及 $(n - k)(T_{k-1} + 1) \ge n - 1$ 即证。

假定 beta、O 同侧活动，另一个 O 必空。按第二空单元是否为 V_v, C_{7v}, C_{7o} 分四类，实际有根森林及未用损失分别给

$$2t_v \ge 43 - w + I + \xi; \quad t_v \ge 14 - w + (\tau - \xi); \quad t_v \ge 76 - w; \quad t_v \ge 126 - w.$$

第一式要求 $26 \le 3w/2 \le 21$；第二式在零洞至少14，在正洞至少32； 后两式也均超过帽形上界。剩余只有 A 中 $\{O_v, B\}, \{O_s, O_t\}, \{O_v, H_v\}$。 全部 alpha、$C_5, U, V, C_7$ 保留，实际所有者给 $2X + G \le 10 + w/2$， 六边森林给 $2X + G \ge 6F - \delta$。因此

$$A_* \ge 368 + I - w/2 > 304 + w \ge A_*, \qquad 64 + I \le 3w/2 \le 21$$

矛盾。这统一闭合 Z_2, P_2。

Z_6 的源等号强制 $I = \tau = \delta = 0$、$\Theta = 160$。 两个首项满容量，$\lambda_v \ge 2M_o + 1$。 由 $J \ge 142, U + J \le 168, C \ge 6$，有 $8M + 4 - S \ge 120$。 两侧固定 C_{14} 之和超过 $8M + 4 > S$，至少一侧四个固定顶标签全 HIGH， 产生正扣减，与 $\delta = 0$ 矛盾。$j > 6$ 的零洞预算直接为负。

4.2 三空与四空的实际接入

四顶尾部与四个完整核心星形给“两帽”必要条件

$$4I+4\tau+\delta+228j\le 538+6w.$$

它排除三空的两帽。四空在整性之前已可由 $2Y>482$ 与 $Y\le 190+3w\le 202$ 排除两帽。 随后两个见证的剩余类恒等式、四边下界 $248-\tau-\delta$、 五边下界 $300-\tau-\delta$ 强制外点整性。 特别是 HIGH 跨中心的计数要保留十个正正固定 HIGH 标签：三空 $C_5\Rightarrow C_6$ 要求 $A_*\ge I+4r+105\ge 235$；四空末链要求 $A_*\ge I+2r+95\ge 161$。 这两行分别超过正洞预算234和156。

$j\ge 3$ 的正洞核心含 $[1,17]$ 中至少12个不同值：六个同时缺值至少花费270质量。 6188个十二值种子的精确子集和检验给中央区间；其他核心点全部能顺序拼接。 故低洞只在 $[1,10]$。每值重数为0、1、2的59049种模式完整枚举给： 四个低洞成本至少256，三个至少195，第一个洞为2至少99。 由此 $P_3:w\le 6,A_*\le 234$，$P_4:w\le 4,A_*\le 156$， $P_5:w=z=2,\tau=67+\sigma,I+\sigma+\delta\le 11$。

三空实际分类只有 A0、A1；A0 的五个实际端点产生十二个重截断模式， 零洞最小余量65，正洞41。A1 的有根并集式给统一下界 391.5， 四个端点模式的零洞上界至多356、正洞至多374。 四空实际分类为 A-E、A-H、B。七个实际端点母式的最小余量为

母式	F	S	E-H	E-alpha自	E-alpha对	E-C5	O
零洞	306	81	253	163	204	295	123
正洞	290	65	237	151	164	283	103

所有由 $e+72h\le A_*$ 允许的重截断标志均纳入，包括两个重端点。 O 母式的常数994从原生端点恒等式和三条森林相加推导，未从旧基数外推。

仅余四空无所有者族 $\{P,B,O_s,O_t\}$。固定标签与截断 H 空隙给相容扣减18。 若两个 H 均命中核心、alpha 下截断均轻，两最大元截断给 $\xi\ge 94+I$。 于是剩余预算至多40；再改用每侧只删一个最大元的同一个 Q， 其两 H 都必须命中 Q。新空隙最多46个标签，使用较弱安全帽 $h_v\le R'+71$， 最终给 $44+2I+12\xi\le 8\tau$，与 $\tau\le 134-I$ 相差至少 $100+22I$。 一重 alpha、一个无核心命中的 H、两个都无命中的三种情形分别由 $100+2I\le 0$、$\tau\ge 91+I>62-I$、$144>134$ 排除。 正洞由五种可能低洞模式给 $\tau-\xi\ge 65$，第一种情形已是 $\xi\ge 86+I$ 与 $\tau\le 138-I$ 的矛盾；其余三种仍有严格余量。

4.3 五空的全部分支

先用首项和两个见证星形证明两个 O 均空，beta 与 H 不能同侧活动； 随后证明整性、跨中心链和同侧链。每个配对的所有者必须来自实际贡献。 零洞预算 $A_*\le 76$。A0 保留 P 的十三种布局由十一张实际仿射端点卡覆盖， 最小余量7；空 P 的 H/C_5 额外空单元给扣减13， 由 $2\tau-2\xi+7F\ge 444$ 得 $483>482$。

alpha 额外空单元必须另证。初始扣减14并不意味着正常性。 先以通用核心命中端点和完整尾部目录得到扣减27；再区分两个物理侧容量， 在同一个实际加权截断上证明

$$|Q|+6F+3\tau-\xi\ge 466,\qquad |Q|+6F+2\tau-\xi\ge 458.$$

第一物理侧给 $B\ge a$ 时上界559，而 $466+4\cdot 27=574>559$，故 $B<a$、扣减31； 固定前缀计数给 $C<b$、扣减35；第二物理侧给 $D\ge a$ 时 $458+3\cdot 35=563>559$，故 $D<a$、扣减38；再五个 LOW 标签给扣减42。 此时才可证明所有保留中心正常。长位移交叠和实际源边缘项使最终运输式给

$$884\le 4I+4\tau-2\xi_s+884\le 542+2(\Theta+B_s+B_t)\le 882,$$

矛盾。把实际边缘项提前放松为20会丢失此矛盾。

A1/B 的九个实际端点运输式分别有最小余量

$$811/2,\ 203/2,\ 258,\ 230,\ 91/2,\ 1105/2,\ 305,\ 415,\ 223.$$

唯一未由这些运输式覆盖的空 P、额外空 C_{5t} 族，使用先前完整的尾部六槽证明， 其相容扣减18给 $\tau+\delta\geq 78>76-I$。该局部定理的整性、唯一 beta 贡献侧、 标签符号和尾部覆盖均在原报告中已证明，本次保持其精确假设。

正洞五空时 $\delta\leq 11$，先由 $K-M-\delta\geq 58$ 排除 O 与 beta/H 活动。 扣减8后剩余至多3，所有原生普通端点正常。A0 空 P 或 split beta 被扣减13或14排除； A1 空 P 的共同11个 HIGH 与18个 LOW 只触17个基线组，扣减12即矛盾。 其余 A0 十一卡最小余量307，A1/B 四个母式余量分别761、326、501、461。 至此零洞和正洞的全部五空情形闭合。

5 覆盖、验证边界与复现

按 $z=0$、$1\leq z\leq 14$、$z\geq 15$ 三段划分：最后一段整体排除； 零洞由 $Z_0,\dots,Z_6$ 及 $j>6$ 覆盖；低正洞由 $P_0,\dots,P_5$ 及 $j\geq 6$ 覆盖。 结构中的低容量非本原、半步长例外在要求本原之前就已排除，因此无循环前提。 对三/四空和五空，验证器各枚举两分支全部 2^{20} 个组件指派， 仅使用已在正文及附录证明的活动蕴含。五空合法指派分为 A0 32、同侧6、空P 10、对侧18、四端点2、split beta 2、B 32、B四端点8。 这证明有限布尔归类无遗漏；它不从有限采样推断实数集合的结构。

在仓库根目录运行：

```
python -I -S -B verification/upper83_complete/check.py --output .work/upper83.json
python -I -S -B -O verification/upper83_complete/check.py --output .work/upper83_O.json
```

入口校验固定源摘要，重新运行原完整84基础、新结构、实际分类、低前缀、正洞、 全秩和五空尾部检查，再核对覆盖及负面控制；不接受输入中的历史 PASS 字段代替计算。 所需 Python 3.10+，无需第三方库、求解器或网络。 写出证明是定理的数学依据；程序验证其有限依赖、整数参数、精确恒等式、完整指派和数值证书。 程序通过不等于用形式化证明助手认证了所有无限论证。

以下附录为与验证器同目录保存的逐项证明说明，包含全部本次新增推导。 其中局部引理写明共同源和保留形式，是证明顺序中的接口；这些前提由正文的结构及分类提供。 旧报告中的“尚未证明83”等历史状态不被改写，本报告在此前局部结果上完成总接入。

A Size84 structural reduction and all high-hole strata

Base cf26c50. These arguments, the common source interface and all local terminals form the size84 exclusion in the companion report. General dependencies: size85 foundation lemmas, source Theta <= 2d + 8, all-rank kernel. Their proofs are cardinality independent; finite size85 rows are NOT imported.

A.1 Fresh parameters and layer identities

n84, d76, e68, m67, a = 33-k, b = 34 + k, I = k(k + 1). Lambda = 1157 + I; Omega = 612-I; C = 2m + 4e + 14-Lambda = -737-I. C + Omega = -125-2I. G = 142. T >= max(48, z, 142-2z), T >= kappa + 142-3z. Unpaired bound2977, odd q <= 53, at least7 opposite pairs. Provisional deficient sign: 1 + T68 = 2347>24*84 + 77 = 2093. Disappearing sign:1 + T67 = 2279>24*76 + 77 = 1901. Uniform rhoP >= -8, rhoQ >= 0 gives k <= 24, a >= 9.

Once common core lattice exists, exact pre-G D1 bound is -125-2I + 2z (+ 16 for a <= 14). Outer comparison Lambda>294 + Omega-rho is strict even with rhoP = -8. For a, b >= 15, maximum removal R <= 71 + rho + kappa + z. Restrict R(E) <= 2R-1, giving fresh: F2-2F1 = -251-2I-c-rho-mu-2kappa-2theta-2ell2-2g; F1-2F0 <= -127-2I + rho-mu + 2z-c-2theta-ell1-g. After actual source layer overlap: C0-U0 >= 126 + 2I + c-2z; C1-V0 >= 250 + 2I + c + mu + rho + 2z; c = 76j + delta. C0 = 1156 + I + tau.

A.2 Small boundaries

A larger AP Q yields z0 and common lattice by residue counting; strict pre-layer comparisons then G142. At a9 Omega12, nonAP Q costs -8 + 53>12, and AP costs T >= 142>20. At a10 Omega60, nonAP Q has kappa <= 16<T10-1; strict inverse, T48 + 52-8>60. AP costs142>68. At a11 Omega106, nonAP Q has kappa <= 63<T11-1. Strict inverse, G142 yields z >= 40. Reuse universal effective witness (r56, beta5, C132), hQ>10 = >muQ >= 189 = >rhoQ >= 125>106 + 8. AP costs142>114. At a12 Omega150: both nonAP, actual sufficient inverse failure gives rhoQ <= 73. Endpoint repair beta3, q14, j19, H6 (see check_structure.py), obtains joint lattice. Pre-layer strict. Then T >= 142-12 and rho >= -8 + 50 imply cost172>150. In actual strict inverse branch kappa <= 108, D1 <= -817. G142 gives z >= 17; effective (55, 4, 119) = >rhoQ >= 113; T48 + 113-8 = 153>150. AP branches handled by general AP argument below (their c caps are negative). At a13 Omega192: failed inverse R94 beta4, q14, j23, H8; cost c <= 192-48-(142-16) = 18<76, w <= 32, j0 obstruction. Strict inverse: if z >= 13, effective (54, 6, 168) givesrhoQ153 and48 + 153-1 = 200>192. Otherwise c <= 26<76, w48, j0. At a14 Omega232: failed inverse R118 beta5, q14, j24, H10 gives c <= 62<76, w40. Strict branch coarse pre-layer strict since kappa <= 184. If z >= 15, independently certified primitive r53 h15 profile gives rhoQ >= 269>232 (no D/G circularity). Otherwise c <= 70<76, w56. j0 in both. So surviving sizes15..52.

A.3 AP branches and low-c lemma

Both-AP branches must be divided at a = 33. For12 <= a <= 32, kappa <= Omega-rhoP<T_b, so residue counting on the larger AP gives one residue. The AP-plus-set formula then has no clipped gap: z = 0, hQ = 0, hP = kappa. This holds for any integer intrinsic step of P; no two-residue bound is needed here. A nonprimitive P is excluded by the low-c argument below. In the only remaining balanced case a33, b34, Omega612<2T33 = 1122. On the common rational scale let p, q be coprime intrinsic AP steps. The two residue counts force p, q <= 2, leaving(1, 1), (1, 2), (2, 1). Unequal steps have z0 and kappa equal to one AP support length minus1, hence at leastT33. Strict pre-G layers and G142 give c <= 612-142-T33<152, w0, and the low-c lemma excludes them. Thus the surviving both-AP case has common unit step. When ONLY the smaller

support is AP, the larger nonAP floor is rhoQ >= b-5. The correct bound is kappa <= Omega-rhoP-(b-5)<2T_a, checked in every row12 <= a <= 33. This is the premise for its at-most-two residue alternatives, including the parity-fiber calculation below. When ONLY larger Q is AP, use rhoP >= n(a), so kappa <= Omega-n(a)<T_b in every row. This licenses the claimed one-residue branch without the false balanced Omega<T_b assertion. For a >= 12, one clipped residue or unique off-scale generator costs T_a + T_(b-2) + 2-2b + rhoP>Omega (rhoP -8 for12, -2 for13, 0 afterwards). Larger AP or one-residue unclipped smaller AP gives z0. Nonprimitive small side cap c <= Omega-142-T_a-rhoP <= 85<152. Since w0, j0/j1 are impossible by already proved zero-hole lemmas; their use only needs common integral source, not individual primitivity. Two-residue smaller AP: normalize U = 2[0, u-1], least odd W, t >= 2 odd Q generators. Exact mass floor muQ + kappa >= u + (b-t)(b-t + 1) + tW + t(t-1)-W-T_b. Against Omega + 2b-2-rhoP, finite minimization forces W <= 17, hence z = W-1 <= 16 and W<u. Exact parity-fiber identity gives hP, hQ <= kappa. The general pre-G D1 comparison (-125-2I + 2z plus small16) is already strict; no special half-step layer constant is needed. G142 gives c <= 93<152, w <= 64. If c<152, either j0 (contradicts w >= 2d + 8 = 160) or j1 with delta <= 75. For j1, w <= 64, K >= 2d-1 = 151 makes leading active. The original one-hole actual owner grammar (using raw first cap12 + eta <= 76, double17 + eta <= 81, and active C7 d >= M + 5 >= 80) gives the standard six-edge terminal and beta/O owner side. The raw caps remain sufficient at w64: single cap12 + eta could76 when eta <= w, so still <80; double <= 81 gives t <= 40.5. Thus same pair arguments hold. Source cap tau <= 380 + w <= 444. All-rank phi_s + phi_t cannot reach42 since min(Tp + Tq, p + q = 42) = 462>444, so F >= 27. Six-edge forest gives2X + G >= 162-delta >= 87. Owner identity and beta/O cap give2X + G <= (17 + w)/2-2 <= 38.5 (use eta <= w; no localization needed). Contradiction. This excludes every half-step branch. IMPORTANT: use eta <= w, not w/2 before localization.

A.4 Both nonAP inverse repairs

Actual sufficient test holds automatically on larger side. Small test floor succeeds k0, 1; failures k2..21 use endpoint choices specified and verified in check_structure.py. Same proof as previous endpoint repair, but q can be any >= 20 satisfying Tq-b-1>R, allowing prefix floor n(q). Fresh main repaired rows k2..18 have H <= 18; diameter margins positive. Pre-G D1 strict. Repaired Q is primitive; if P nonprimitive its cap c <= 65<76, w <= 72<160, so j0 obstruction excludes. Thus P primitive. For STRICT inverse main branches, z <= 62 + I makes D1 strict. If z >= 63 + I, both cores individually primitive already (strict inverse hole bound). We can use the UNIVERSAL dense h49 certificate BEFORE D-integrality; it does not depend on G142. If one final support splits disjointly, it costs rhoX >= T_(r-2) + 2-2r, while other hasrho >= profile49(s). Across all38 oriented sizes15..52, cost + profile49(s) + (63 + I)-Omega >= 77. Hence both final translates intersect. Maximum-removal hX/2 sharpening gives D1 <= -125-2I<0. This replaces all large-hole endpoint repairs and is noncircular. Original source outer layer then aligns and intersects.

A.5 Complete high-hole stratum

With sizes15..52 and primitive cores, paired universal floors over a + b67: h15: (mu, rho_elementary) = (418, 288); h17: (461, 331); h21: (503, 373); h49: (539, 409). Use independently verified profile table (all relevant r15..52 present). Two-max source gives J + U <= 624 + w-I-tau-c. Outer layer gives2L <= 1530 + w-2I-2c-mu-rho-2z. Then M <= 836 + H + (rho-mu)/2 + z-I-c; core central width W0 = 1156 + I + mu-H satisfies W0-M >= -904 + 4I + 3c + 3(mu + rho)/2-3z. Bands15..16, 17..20, 21..48 give positive margins107, 224, 266. Their I + c caps214, 179, 191 and H caps246, 219, 289 give individual core mass minusH minima139, 166, 96. Tailz >= 49: kappa <= 203-I-c, H <= 406-2I-2c, I + c <= 154, so each coremass>H (compute minimum). If j >= 1, c >= 76, then z <= 127-I-(c-76), width margin >= 518 + 3c-3z >= 365 + 3I + 6(c-76)>0. If j0, use unlocalized universal raw bound w >= 2M + 10 and w <= 4z to getM <= 2z-5

<= 401; W0 >= mu + 2rho-68 + 3I + 2c >= 1289. Hence same strip propagates in all cases, w <= z and even. For j >= 1 source A <= 380 + w but tau >= 418 + 2z, contradiction (stronger in later bands). For j0 and z <= 48, w <= 48 contradicts no-empty w >= 160; in tail tau >= 539 + 2z and A <= 456 + w <= 456 + z contradict. All high-hole families gone without old first-gap H1 transport.

B Low-hole localization and the one-hole terminal

The common integral primitive source and1 <= z <= 14 are hypotheses. The common source hypotheses are supplied by structure.md.

B.1 A simpler localization, valid for every j >= 1

Source bounds hold uniformly: A <= 456-76j + w, e_v <= 621-76j + w-A, w <= 4z <= 56. Positive cores have mu >= 61. Set H = tau-mu. Since min_r(T_r + I) = 385 and max_r(r-I) = 34,

```
C_v-H>=76j-w-10+delta>=10+delta,
W0=C0-2H>=822+76j-w+2I+delta,
M_v=R_v+e_v<=655-76j+w-I-delta.
```

At j1, w56, W0 >= 842 and every one of the eight peeled source entries is <= 635. Thus each translated central interval joins as each source entry is adjoined, without using a large coarse layer gap. All j >= 1 satisfy stronger bounds. The common-strip theorem gives w, z even, w <= z <= 14, equal target hole counts and boundary localization. This replaces the more complicated j >= 3 localization in positive.md and also supplies P1/P2.

B.2 P1: reuse the sharp one-hole terminal

With one ordinary whole hole, the active own centres determine all outer coordinates integrally without requiring leading activity. Each sign has an active C6 or C7, so d_v >= M + 5 >= 80; moreover its leading maximum is positive. Thus the leading cardinality/loss inequality and raw-gap lower guard are licensed even if its full leading resource has no target hit (its entire K loss is still in delta). Its shift is already integral.

If P were empty, an actual beta contributor and the opposite C5 give t_opposite>d_beta >= 80, contradicting the active O cap <= 19. Thus P has a contributor v, forcing t_v>d_o >= 80 and making O_v the unique hole. Its own beta cannot contribute, because H_v is active and beta/H would give t_v <= 16. Hence branch A has sole beta on u = o, with beta_u, H_u, O_u active and all alpha, C5, U, V, C7 retained. The identical six-edge rooted proof from the zero-hole one-hole lemma gives

```
2t_u>=I+max(99-5xi+tau,27+2xi)-w>=51+I-w.
```

The double raw cap gives2t_u <= 17 + w/2. Their difference is at least34-3w/2 >= 13. This closes P1 without first-gap or two-cap subcases. Source bounds guarantee at least14 core entries on each side, so the actual six-rank trim used in the displayed max exists.

B.3 P2

The former long first-gap route has been replaced by the shorter complete shared proof in z2_p2.md. It uses a32-unit prefix defect and the same four beta/O cards as Z2. No eleven-value seed or P2 first-gap census is needed.

C A shared two-hole proof: Z2 and P2 at size84

Common source hypotheses: d76, m68, r_s + r_t67, complete targets (Z2) or localized positive targets with even w <= 14 and eta_v = w/2 (P2). A = I + tau + delta <= 304 + w; R = 67 + xi; K-M >= R-302-w + A. The source max bounds are e_v <= 469 + w-A and M_v <= 503 + w <= 517 after localization (the larger559 initial bound is also valid). Every integral ordinary centre in[1-M_v, S_v] is active; the P2 proof of this property is supplied below. For Z2 it follows simply from the actual source points0, M_v and S_v >= M_v.

C.1 Restore leading activity in Z2

If leading v is empty, delta >= K and the source inequality K-delta >= I + tau-8 gives I + tau <= 8, 31 <= r_v <= 36 and u = delta-K <= 8-I-tau. Both leadings cannot be empty because K >= 151>u. The opposite leading has an integral shift. If at least two of the opposite four own centres are holes, all four own centres on v are retained and determine integral v-coordinates. Otherwise three opposite own centres together with its integral leading shift determine all opposite coordinates. Then an active C6_v or C7_v gives integral lambda_v; if both are holes all four own centres on v are retained. Thus lambda_v is integral in all cases. The usual two-witness residue argument, now with both leading shifts integral, gives all outer coordinates integral; it does not need a hit in both leadings.

Every other retained singleton has native loss <= u <= 8, so is normal: a BAD centre has at most max(r_s, r_t) + 4 <= 40 hits and loss >= 36. Since A >= K >= 151, e_v <= 318. Thus S_v-2M_v >= T31 + 2*31 + 6-318 = 246>0. A leading centre in [1-2M_v, S_v] would hit via0 or2M_v. If lambda_v>S_v, at least two of its four own centres remain retained and HIGH, each losing r_v + 5>u points. Hence lambda_v <= -2M_v.

All four cross centres on v are then LOW. In A this removes three cross singletons, impossible. In B the two exact holes must be C6_v, C7_v; both paired resources have only the opposite contributor. All own centres are retained. Normal V_v has V_v >= 72-r_v-u. But V_v = lambda_v + v4 <= X_v-M_v, so X_v >= M_v + 72-r_v-u >= 76-u >= 68. Active C7_o gives t_o >= 2X_o + X_v + g_o + 1>69, contradicting active O_o' s cap12. Thus both Z2 leadings are active. The same residue argument proves outer integrality without any exceptional leading cases remaining.

C.2 P2 activity and a minimal positive-core defect

The common low-hole localization gives A <= 318 and18 <= r_v <= 49. K-delta >= -8-w + I + tau>0 using tau >= 61 + 2z and w <= z. Thus both leading shifts are integral and the two-witness argument gives outer integrality. Source side floors R_v >= r_v + 1 and tau_v >= r_v-3 imply

```
S_v-H-R_o>=T_r+7r-606-2w+2I+2delta>=10+2delta>0.
```

The quadratic minimum before subtracting606 + 2w is644. A target-hole centre has r_o >= 18 distinct opposite-core images in the positive hull, more than eta_v <= 7 missing targets. Nonholes use zero; nonpositive centres use M_v below the boundary strip, since S_v-M_v >= C_v>H. This proves the activity interval used above.

Let f be the first missing absolute-core subset sum. The greedy rule gives f<max core <= 352, since r_v-I <= 34 and I + tau <= 318. The side with at least34 core entries has a prefix obtained by deleting its maximum, of mass at least T33 = 561. That prefix still misses f, hence its subset sums are not a complete interval. For n distinct positive integers with an incomplete subset-sum interval, their mass excess over T_n is at least n-1. Indeed, if the first greedy failure follows k entries, all n-k subsequent rank deviations are at least T_(k-1) + 1 for k >= 1; their sum is at least n-1. For k0 the bound is n. The elementary inequality (n-k)(T_(k-1) + 1) >= n-1 holds for1 <= k<n, first at n = k + 1 and then increases in n. Applied to this prefix, it proves

tau-xi>=32 (positive case only).

C.3 No two caps

The common four-top and star calculation gives 4I + 4tau + delta <= 82 + 6w if both cap predicates hold. For P2 this contradicts tau >= 61 + 2z, w <= z directly. For Z2 it gives tau + delta <= 82. If any C7_v is active, the valid four-edge arm gives t_v >= 249-tau-delta >= 167>12. If both C7' s are empty, they are the exact holes and their lower C6' s are retained. Both C7' s are HIGH, so the leading losses total at least r_s + r_t + 160 = 227>82. Therefore no two caps exist in either case. At least one O is empty. Here a cap means active O or active beta, H.

C.4 Four exact beta/O exclusions

Suppose beta_v, O_v are active. The sharp double cap says 2t_v <= 17 + w/2; the opposite O is the first hole. Let W be the other whole hole. All listed native resources below are original, distinct retained resources; every leading is kept at its full K capacity. Roots explicitly retain their native loss in the forest.

1. W is none of V_v, C7_v, C7_o. Choose a full physical-v two-edge arm and the first edge of the other, rooted at leading v. The unused C7_v has upper image t_v-2X_v-X_o-g_v <= t_v-4, hence native loss >= 80-t_v. This quantity is <= 76 whenever C7_v is active; otherwise the assumed case is already impossible. The rooted forest gives t_v >= K-M + 3F-delta + 80-t_v. Since tau + 3F >= 198, this implies 2t_v >= 43-w + I + xi. It contradicts the double cap because 26 + I + xi <= 3w/2 <= 21 would be necessary.
2. W = V_v. It is HIGH, since active leading implies V_v>1-M_v. Active beta gives alpha_v>U_v>V_v. Thus the three retained alpha_v, H_v, O_v each lose at least r_v + 5. Use shortcut edges leading_v–U_v and leading_o–C7_o, root leading v, and the three unused losses. The full-core overlap F0 = 68-xi gives t_v >= -84-w + I + tau-xi + 3r_v >= 14-w + (tau-xi). In Z2 this is >= 14>8.5. In P2 it is >= 46-w >= 32>12 by the just-proved defect32 lemma.
3. W = C7_v. It is HIGH above retained C6_v. Use the retained three-edge forest leading_v–V_v and leading_o–C7_o–C6_o, root leading v. The three unused U_v, H_v, O_v lose at least r_v + 5 each. Hence t_v >= -220-w + I + tau + xi + 3F + 3r_v >= 76-w>12, using tau + 3F >= 198 and I + 3r_v >= 98.
4. W = C7_o. It is HIGH above C6_o; its leading loses r_o + 80 and the three unused V_o, U_o, H_o lose r_o + 5 each. Use edges leading_v–V_v and leading_v–U_v with both leading roots (the opposite is isolated). The total gap is2X_v + g_v. The full-core row gives t_v >= -4-w + I + tau-xi + 4r_o >= 126-w>12, since I + 4r_o >= 130.

Thus simultaneous beta and O activity is impossible. The one-empty arm rule is used only where it has at most one empty vertex per separated arm; no capacity is restored to such a vertex.

C.5 Exhaustive remaining holes and a common terminal

If both O' s are holes, all other whole units are retained; B would give both beta/H caps, so the branch is A and beta has a sole owner. If only O_v is missing, beta_o cannot contribute because O_o is active. In B its singleton is the second hole, leaving beta_v, H_v active and two caps. In A, either the entire beta pair is the second hole or its sole contributor is v and H_v must be the second hole. This exhausts exactly {O_v, Bpair}, {O_s, O_t}, {O_v, H_v} in A. All alpha, C5, U, V, C7 singletons are retained. Choose the cap side u opposite the single missing O, or the

sole beta owner if both O’ s are missing. Actual alpha_o, C5_u activity gives t_u >= 2X + G + 2, hence2X + G <= 10 + w/2.

The two six-edge retained trees on one F-optimal actual core give 2X + G >= 6F-delta. The universal full-rank kernel tau + 6F >= 378 then forces A >= 368 + I-w/2. Comparing A <= 304 + w yields

```
64+I<=3w/2<=21,
```

a contradiction. At w0 the margin is64. This closes Z2 and P2 under their common-source hypotheses, without a first-gap table.

D Positive-hole activity and terminal proofs

Common integral primitive source and 1 <= z <= 14 are supplied by structure.md. The uniform localization in localization_p12.md gives even w <= z <= 14 and eta_v = w/2 for every j >= 1. The exact finite replays are in check_positive.py. P0 is excluded by the unlocalized no-empty bound; positive j >= 6 is impossible from A <= w <= 56<tau.

D.1 One dense seed and a ternary prefix census

After localization and j >= 3, A <= 242, C0 <= 1398, r_min >= 21, and each core entry <= 276. Six values missing from both sides in[1, 17] cost tau >= 6*67 + 42-2*(12 + 13 + 14 + 15 + 16 + 17) = 270>242. Thus at least twelve distinct values occur there. The retained exact6188-seed lemma gives [11, T_B-11] in FS(B) and at most five lower holes. Adjoin the other labelled core occurrences in sorted order. For a current n >= 34, its total is at least2T17 = 306, so x <= 276 <= T-21. For n <= 33 a failed join x>T-21 implies x >= 58 and at least34 remaining entries, forcing C0 >= 78 + 34*58 = 2050>1398. Thus every join succeeds and all lower holes are in[1, 10].

The exact59049-pattern census enumerates multiplicities0, 1, 2 at 1, …, 10, then fills the remaining67-count slots from11 upwards, at multiplicity at most2. It rigorously lower-bounds I + tau without requiring an assignment to physical sides. Four lower holes cost at least256; three cost at least195; the first gap2 costs at least99. Consequently:

- j3: z <= 6, w <= 6, A <= 234.
- j4: z <= 4, w <= 4, A <= 156; possible lower-hole sets are precisely a subset of { {1}, {1, 2}, {1, 3}, {2}, {3} }.
- j5: the only lower hole is1. Hence z = w = 2, all positive core entries omit1, tau = 67 + sigma, and I + sigma + delta <= 11.

For j4, tau-xi >= 65. At first gap1 every rank deviation is >= 1, so removing the two maxima leaves at least65 mass. At first gap2 the low multiset is {1}; the other entries exceed2, giving an even larger bound. At first gap3 it is {1, 1}; the remaining entries exceed3, again giving at least65. This proves the bound directly for every listed profile.

D.2 Activity interval (j >= 3)

The bounds above give S_v-H-R_o >= T_r + 5r-480 + 2I + 2delta >= 113 + 2delta. Also S_v-H-M_v>0 from the source side mass. A target-hole centre in[S_v-H, S_v] has at least r_o >= 21 distinct opposite-core images strictly inside the target hull, more than eta_v <= 7 holes. Thus it is active. Other centres in[1, S_v] use zero, and centres in [1-M_v, 0] use M_v in the complete lower target strip. Therefore every integral centre in[1-M_v, S_v] is active. An empty centre above an active one is HIGH>S_v. This supplies the positive-case activity rules used by the zero-case classifiers without assuming complete targets.

D.3 Fresh two-cap identity and j3/j4 grammar

The four-top tail has at most4*(76 + 6) + 5 = 333 values per sign: d_s + d_t + G + delta-2K >= 786-76j. The four full-core stars give 2X + 4R + delta >= 540. With both caps t_v <= 12 + eta_v, put Y = 2M-2K + 24 + w + delta. Then G + 2X <= Y, G >= 786-76j-Y, and Y <= 932-152j + 3w-2R-2I-2tau-delta. Hence, whenever the stars are licensed,

```
4I+4tau+delta+228j<=538+6w.
```

This excludes j3 for w <= 6. For j4, before integrality or the stars, the first two inequalities give2Y>482, but Y <= 190 + 3w <= 202, so two cap predicates are already impossible.

The cardinality-independent residue classifier in Appendix D now works with fresh strict arm bounds. For j3 the two-witness argument immediately gives integrality. For j4, two nonintegral lower coordinates use four holes and leave both O active. With just one nonintegral lower coordinate of residue theta and top residue b, the five residues H, O, alpha, beta, C5_opposite are b, b + theta, b-theta, b-2theta, -b. If b! = 0 the actual branch-A/B hole counts again leave both O active. If b = 0 the exact holes are the two witnesses, O_v, alpha_v. The original four integral-gap aggregate edges give at least248-tau-delta >= 96-w >= 92; either active opposite O, C6 or beta, O, C7 contradicts caps <= 14. A nonintegral top pair after lower integrality costs four holes and the five-edge bound300-tau-delta >= 148-w >= 144 contradicts the retained opposite O, C6. Thus all outer coordinates are integral.

The separated-arm pointwise rule allows one empty vertex per arm with zero capacity; failure to find an arm at j4 is exactly the four-hole witness configuration already excluded. Therefore active C6 forces t >= 301-tau-delta, and active C7 forces t >= 249-tau-delta. At j3, w <= 6 these are at least67 and15; at j4, w <= 4 they are at least145 and93. They exclude O/C6, beta/H/C6 and beta/O/C7 simultaneous activity.

A HIGH cross makes leading loss at least r_v + 80. At j3, empty C6 above active C5 leaves at least three higher native resources, each losing r_v + 5. The ten positive-positive fixed labels are HIGH and compatible with all baseline groups. Thus A >= I + 4r + 105 >= 235, contradicting A <= 234. At j4 the C4 = >C5 and C5 = >C6 implications have A >= I + 3r + 90 >= 188>156. C6 = >C7 uses one higher resource and the ten HIGH labels: A >= I + 2r + 95 >= 161>156.

The literal owner identities and hole counts therefore give the same complete j3 A0/A1 and j4 A-E/A-H/B tables as Appendix D. For j4' s beta/O exclusion, the three-edge rooted arm when V_v is retained gives t >= 119-w + I + xi; if V_v is missing, its HIGH status and three unused higher native losses give t >= 68-w + I + 3r_v. Both exceed the sharp double cap. Consequently both O are holes. The remaining two-beta-hole B alternative has necessary row 2T_r + 9r + 4I + 7tau/4 + 2tau_v <= 1104 + 4w <= 1120, while its integer quadratic minimum is1160. It follows from the actual HIGH-beta master2S_v + K + 3g_v + 6X_v <= M_o + 1147 + w + delta and the three-edge bound186-3tau/4-delta. The owner-side HIGH beta contradicts 2d_v <= M_o-1 and actual P activity d_v>M. Thus neither extra beta-hole alternative is omitted.

D.4 j3 terminal bridge

The A0 geometric master in z34_cards.md only changes1026 to1026 + w. Use the relaxed common domain A <= 234, L >= max(8, A-82), max(4, A-162) <= e_v <= 399-A, and keep the heavy thresholds70/72. The light-rank product276 still exceeds tau + eps <= 226. The exact twelve minima at w6 are 162, 224, 285, 330, 126, 266, 422, 88, 158, 313, 41, 220, all positive.

For A1, the same actual side-union argument gives47 + w in place of47. Its four Z upper bounds become212 + 3w, 288 + 3w, A + 52 + 2w, A + 128 + 2w. Their largest value is374, below Z >= 391.5. Thus all j3 positive-hole forms are excluded without individual first-gap terminal cards.

D.5 j4 terminal bridge

Each of the seven Z4 actual masters in z34_cards.md gains at most w on its right. Use A <= 156, L >= max(8, A-4), max(4, A-84) <= e_v <= 321-A. Include every heavy flag allowed by e + 72h <= 156 (even two in Card O if needed). The seven finite minima are290, 65, 237, 151, 164, 283, 103, all positive.

For the exact ownerless family use z4_ownerless.md with target hole allowances. The initial four-arm bound is116 + I-w; hence X + g_v >= 57. Use the stronger active-C5 identity to

obtain d_v-h_v-X_v >= X + G + h_o-55-eta_v>0 even at w4. Thus the ten-label gap yields h_v <= M + 11 + eta_v. The same e18 ledger works: V2_s + V2_t <= 20 + w-3X-G<0. The core gap has at most30 + eta_v positions, so h_v <= R + 31 + eta_v with a core hit, or h_v <= R_v + M_o + 30 + eta_v otherwise. Both H core-hit, light alphas imply xi >= 94 + I-2w >= 86 + I. But the positive profile has tau-xi >= 65 and tau <= 134 + w-I <= 138-I, contradiction. One heavy alpha gives100 + 2I <= 3w <= 12. One H with no core hit gives tau >= 19 + u + I-2w >= 83 + I, while delta >= 90 gives tau <= 66-I. Two non-core-hit H losses144 exceed eps <= 138. Thus the ownerless positive family is excluded as well.

D.6 j5 complete terminal

The prefix census leaves just lower hole1. Thus z = w = 2, tau = 67 + sigma and I + sigma + delta <= 11. All core entries omit1: xi_s = 1 + xi' _s, xi_t = 1 + xi' _t, xi' _s + xi' _t <= sigma. In particular r_s, r_t are between31 and36. Before outer integrality, K-M-delta >= R-76 + I + tau >= 58. The leading inequality rules out active O and active beta/H because their raw caps are13 and10. The residue argument and literal owner identities of z5.md now apply. A HIGH cross loses r_v + 80>delta leading points, supplying the same cross chains. Both O provide fixed credit8. Every retained singleton or actual sole pair is normal: a BAD centre loses at least72-max(r_s, r_t) >= 36, while eps <= 3. In particular all own chains and actual endpoint lists are licensed.

The source rows are freshly

```
S=2K+1074-delta, A=I+67+sigma+delta<=78,
L>=A+74, max(4,A-6)<=e_v<=243-A.
```

Do not import the zero-hole strengthened bound L >= A + 78. Use the actual full core q = 68, F = 68-xi_s-xi_t >= 63. For A0 with empty P, the fixed ledgers for H/C5 extras have credit13 and the alpha extra has credit14, each exceedingdelta11. Split beta has both H HIGH and credit14, also impossible.

For A1 with empty P, the shared HIGH11 ledger from O_s, O_t, H_s combines with these18 LOW labels: C55, C56, C57, C58 on both signs; V5, V6, V7, V8 on the inactive-beta sign t; F3, F4, F5 on both signs. They touch17 baseline groups, so credit = 11 + 18-17 = 12>delta. Pair1 is LOW, since a HIGH component would lose r_v + 80>delta leading points. Beta_t is LOW because one of H_t, alpha_t is retained and normal. The F signs follow from d_v>g_v. These signs hold for every possible final companion hole. No inactive paired component is assigned native capacity.

Retained-P A0 uses the same eleven exact affine cards as z5.md, with normal endpoint errors zero. Their full-core minima, in A–K order, are608, 452, 723, 428, 344, 504, 504, 307, 515, 597, 504. Card E uses credit11 from HIGH H_t; others usecredit8. Card D uses both actual triples on this same full core, giving5q + 6F. For remaining A1/B, use opposite/same/four-endpoint/branch-B physical masters from z5.md with their right side increased by2. Even retaining their old nonnegative xi error allowances and omitting the nonnegative side-mass terms, the minima are 761, 326, 501, 461. check_positive.py enumerates the entire integer domain I + sigma + delta <= 11, xi_v >= 1, xi_s + xi_t <= 2 + sigma and the full peel polygon. This closes every P5 form.

E Fresh geometric derivations for Z3/Z4 terminal cards

These are local terminals, assuming the actual retention forms and common integral source. The classifiers in positive.md, z5.md and check_geometry.py provide the actual retention predicates separately from these scalar terminals.

E.1 Z3 A0

S = 2K + 1224-delta, A <= 228, L >= A-76, max(4, A-156) <= e_v <= 393-A. The five weighted endpoints are P_s, 2alpha_s, alpha_t, C5_t, H_s. Their literal identity is H_s = P_s + 2alpha_s + alpha_t + C5_t + 4(M + X) + 2G. The endpoint constant is5*72 + 71-3*67 = 230. The residual physical source mass is at least T_r + 4r + tau_t + 4xi_t + 10. With K <= M + R + 21 and the six-edge retained forest, the exact master is

```
T_r+4r+tau_t+2L+4xi_t+12F<=1026-2e+delta+B_*.
```

The common grouping rebates every endpoint's unused native loss except the algebraically doubled alpha_s. Consequently B_* = 2b_alpha_s + sum_(other heavy Z)(b_Z-k_Z). Call other endpoints heavy at clip70; alpha_s's flag a is reserved for a peeled clip >= 72. If a = 0 its error <= xi_s. All other light errors satisfy b <= k: outside-hull clips have24 <= k <= 69, and k(73-k) >= 276>220 >= tau + k. The next-rank inequality (72-k)b <= tau proves the claim. Normal endpoints have error0. Every heavy error is at most xi_v + e_v-4, including core ranks70, 71. When a = 1, alpha_s <= 0 and the eleven LOW labels U5:8_s, V5:8_s, F3:5_s touch nine groups. With the eight O HIGHs, this gives compatible e10. Otherwise use e8.

Thus with n_s in{0, 1}, n_t in{0, 1, 2, 3},

```
B_*<=2xi_s+(2a+n_s)(e_s-4)+n_t(e_t-4)
      +n_s xi_s+n_t xi_t-70(n_s+n_t),
delta>=e+72a+70(n_s+n_t).
```

These are precisely the twelve scalar systems exhaustively checked by the full-rank verifier, with minimum margin65. The actual five resources and the six-edge forest have been specified; no additional capacity is assigned to an inactive component.

E.2 Z3 A1

The actual holes are O_s, O_t, H_s; beta_s and P_t are sole resources. The same twelve HIGH labels as in the former physical-union proof give e12. Group leading s and its seven native resources: S_s >= K + 532-eps. Lower P_t and upper H_t are the only endpoints. Let B = Gamma_Pt + Gamma_beta_s and H = Gamma_Ht <= 75. The ten-edge forest and the two X_s star edges give, on one Q, Qgap + X_s >= 6F-eps + B + H/2. The identity H_t = H_s + P_t + 2s4 + t4 + g_s + g_t + h_t and HIGH H_s imply S_t >= S_s + M + M_s + Qgap + X_s + 77-Eclip-Eerr. As Eclip <= B + H, the source sum1224 gives exactly

```
M+M_s+6F-H/2<=47+2delta+Eerr.
```

The four endpoint flags p, h have Eerr <= xi_s + xi_t + p(e_t-4) + h(e_s-4). Define Z = r_s + 2I + 2tau + xi_s + 6F-H/2. The source peel bounds give, respectively, Z <= 212, 288, A + 52 <= 280, A + 128 <= 356. But2tau + 6F >= 396, r_s + 2I >= 33 and H <= 75 give Z >= 783/2 = 391.5. All endpoint positions are excluded with margin at least71/2.

E.3 Z4 physical capacities and masters

S = 2K + 1148-delta, A <= 152, L >= A, max(4, A-80) <= e_v <= 317-A. The generic rank constants are72-r_v and S_v + r_o-71. A native clip<72 has error <= xi_v; a heavier one has error <= xi_v + e_v-4 and costs at least72. All heavy flags allowed by e + 72h <= 152 must be included, in particular two flags when e8. No unproved assertion that at most one is heavy is used.

The original endpoint/forest lists for F, S, EH, EA-own, EA-opp, EC5 are retained literally. Physical t with seven native resources gives S_s <= K + 616-e; six gives S_s <= K + 692-e. These are union-deficiency bounds, not claims that their images are disjoint. Direct substitution of the endpoint identities gives these fresh masters (E is the actual weighted error, N the listed endpoints' native loss):

- F: T_r + 4r + tau_t + 2L + 4xi_t + 12F <= 950 + delta-2e + E. The nominal constant is230. The six-edge forest avoids all five endpoints and Eclip <= 2N. There may be one empty ordinary vertex, charged by the separated-arm one-empty rule, with zero capacity.
- S: 2M + 9F + 4e + 2N <= 539 + 3delta + E. Nominal constant153; physical t has six resources. The leading-t, six-edge and three-edge forest budgets rebate all four endpoints.
- EH: R + 2L + xi_t + e_t + 10F + 3e + N-h <= 412 + 2delta + E, where h is the upper V_s deficiency. Nominal constant225-r_t; physical t has seven resources. The six-edge forest avoids all endpoints, while the four stars contain V_s only. Drop N-h >= 0.
- EA-own: L + 2e_t + 2xi_t + 10F + 3e + N <= 415 + 2delta + E. Use alpha_s >= 1-a and physical t's seven resources. All four endpoints avoid both the eight-edge and two-star systems.
- EA-opp: R + 2L + xi_s + e_s + 11F + 3e <= 487 + 2delta + E. Use alpha_t >= 1-b; nominal constant159 + r_t; physical t has six resources. The six-/five-edge forests avoid all weighted endpoints.
- EC5: R + 2L + xi_t + e_t + 11F + 3e + N <= 488 + 2delta + E. Nominal constant225-r_t; physical t has six resources. Its seven-/four-edge forests avoid the five endpoints.

Subtract baseline rank errors and each actual heavy increment e_v-4. Use N >= 72h where it occurs. The resulting systems are exactly the six first Z4 cards in the full-rank verifier. Their margins are 306, 81, 253, 163, 204, 295.

E.4 Z4 O: deriving the previously conditional constant994

Its predicate is both O and H_s empty, beta_s native and P_t sole, all eight lower own/cross resources retained. The twelve HIGH labels give e12. Put B = Gamma_beta_s + Gamma_Pt, H = Gamma_Vt. The actual three forest rows are

```
X_s+g_s>=2F-(eps-B-H)/2,
X_s>=F-(eps-B-H)/2,
X_s+X_t>=2F-(eps-B)/2.
```

Three, two and one copies give 6X_s + X_t + 3g_s >= 10F-3eps + 3B + 5H/2. The endpoint identity V_t = 2d_s + beta_s + P_t + 2a + b + g_s, with twice leading s, has Eclip + 2ell_s <= 2eps-(B + H). Consequently

```
S_s+3g_s+4X_s+2a+b
<=1000+r_s+delta-2e-B-H+Eerr.
```

Insert S_s >= T_r + tau_s + 4R_s + 6 + e_s and the forest row. This yields

```
T_r+4r+5xi_s+tau_s+R+L+2e_s+10F+60+4B+7H/2
<=994+4delta+Eerr.
```

Thus994 is derived from the actual physical resources, rather than extrapolated from1006. Baseline Eerr <= 2xi_s + xi_t. A heavy beta_s or P_t costs4*72 = 288, a heavy upper V_t costs(7/2)*72 = 252; subtract its source e_v-4. This is exactly the seventh full-rank Z4 system, with verified margin123.

The exact ownerless family, to which none of these cards applies, has its separate proof in z4_ownerless.md. The preceding activity lemmas and the independent component census supply the exhaustive size84 classification; terminal arithmetic alone would not do so.

F Size84 Z4 ownerless terminal

Hypotheses: common integral source, complete targets, exact holes {P, Bpair, O_s, O_t}, all fourteen singletons H, U, V, C5, C6, C7, alpha retained, and integral outer coordinates. A = I + tau + delta <= 152, R = 67 + xi, L >= A, e_v >= max(4, A-80), e_v <= 317-A. Neither empty pair is assigned a contributor.

F.1 Actual fixed credit and first H gaps

The eight O HIGH labels give eps8 = delta-8. The four-edge arm on an actual F-optimal core, with tau + 4F >= 260, gives 2X_v + X_o + g_v >= 4F-eps8 >= 116 + I. Thus active C7_v gives d_v >= M + 117 + I and X + g_v >= 59. Active alpha gives h_v >= X_v + 1. The original raw55 gap first gives h_v <= M + 56= X + g_v-55 >= 4. Consequently exactly the same ten fixed candidates can enter (U_v + M_v, H_v-M_o): C33, C34, C44, C15, C26, C27, C28, U2, U3, V2. Their complete sign audit in Appendix D is valid with these strict margins. It gives h_v <= M + 11. Both empty paired centres are LOW. The thirty-two LOW labels listed in that appendix touch25 groups; one witness from each reflected pair V2, C35, C45 adds three LOWs in already touched groups. For V2 use20-3X-G<0 since3X + G >= 232 + 2I. The untouched C25 group contributes another LOW witness. With eight HIGHs this gives44 misses against26 baseline groups: e = 18.

The original core-gap list has37 candidates. Its initial bound h_v <= R_v + M_o + 37 removes all alpha peel contributions because d_v-h_v-e_v >= 80 + I>0. Also a-U3 >= X + g_o + h_o-36>0, where a = U_v + R_v. It removes U3, U4, V2. The remaining30 candidates yield

```
h_v<=R+31 if H_v has a core hit,
h_v<=R_v+M_o+30 otherwise.
```

In the second case the72 source points at or above -R_o miss above the complete target, so H_v' s native upper loss is at least72. The whole-side source bound also gives S_v>M: directly S_v-M >= T_r + 4r-378 + I + delta >= 93 + delta>0. This licenses the interval/core-hit statements without hull gaps.

Let N be the sum of the two alpha deficiencies, k their lower clips, ell the sum of the leading losses, and B the sum of the alpha rank errors. Then alpha_s + alpha_t >= 77-k-B, N >= k. Write eps = delta-18. For any actual common core with overlap f, the rooted stars give

```
2X_v>=2f+N+ell-eps,   2X>=4f+N+ell-eps.
```

An unused H loss u can additionally be rebated from the second row. For light alpha clips (<72), B <= xi. At most one clip is heavy, since eps <= 134<144.

F.2 Both H have core hits, both alpha clips light

The H caps and alpha endpoints give L + X <= R-15 + k + B. Combine the stars, L >= A, N >= k and k <= eps to obtain

```
4f+2I+2tau+36+ell<=104+4xi.
```

Delete two maxima on each core side: q = 64, f2 = 64-(tau-xi)/2. Its substitution forces xi >= 94 + I. Thus tau >= 94 + I, delta <= 58-2I, eps <= 40-2I.

Now delete just ONE maximum on each core side. Let the remaining maxima be P' _v = r_v-1 + zeta_v. Put zeta = zeta_s + zeta_t <= tau-xi. This one actual Q has66 points, span R' = 65 + zeta, and overlap f' = 66-zeta. Every H has a Q hit: otherwise at most the ten removed source entries can hit, forcing native loss at least66>eps. Every alpha clip is <= 40<71. Its next

rank is strictly below the core maximum, so its rank error is bounded by zeta_v; hence B <= zeta.

For completeness the Q-gap count is bounded independently. In (U_v + P’ _v, H_v-P’ _o), only five positive non-Q entries from each of U, V, alpha, five negative non-Q entries from H, and at most15 leading pair labels can enter. Cross translates lie wholly below U_v. A leading pair above U_v + P’ _v must use two of the five positive entries above P’ _v. The same eleven fixed labels from the original truncated gap list suffice: its exclusions use only P’ _v>0, d_v>h_v, d_v-o2 + P’ _o>0 and the already established outer order. Thus there are at most15 + 5 + 15 + 11 = 46 labels. In particular the deliberately weaker safe cap h_v <= R’ + 71 holds.

The two new H caps and the improved alpha errors imply

```
L+X<=2R'-R+65+k+zeta.
```

Use the same rooted stars on Q and eliminate N, k as before. This gives2I + 2tau + 2xi + 264 <= 220 + 10zeta, hence

```
44+2I+12xi<=8tau.
```

But xi >= 94 + I and tau <= 134-I make the left side minus right side at least100 + 22I>0. This excludes the light/light case.

F.3 Other H/alpha statuses

If both H have core hits and alpha_v has clip k_v >= 72, the opposite alpha endpoint and H cap give e_o + X_o <= 26 + xi_v + k_o + zeta_o, with zeta_o <= xi_o. Use the individual rooted star on the one-trim support with f1 = 66-tau + xi, N >= k_v + k_o and k_o <= eps-k_v. Then

```
2e_o+260+ell<=2(tau+delta)<=2A-2I.
```

Since e_o >= A-80, this would give100 + 2I + ell <= 0. The heavy alpha’ s unbounded peeled error was never estimated.

If exactly one H has no core hit, its unused loss u >= 72 leaves N <= eps-u <= 62, so both alpha clips are light. Its truncated cap and the other H cap give e_v + X <= R-16 + k + B. Rebate u in the rooted stars, use k <= eps-u and f1 = 66-tau + xi, obtaining

```
2e_v+198+2u+ell<=4tau+2delta.
```

With e_v >= A-80 this forces tau >= 19 + u + I >= 91 + I. But delta >= 18 + u >= 90 gives tau <= 62-I, contradiction. If neither H has a core hit, their two losses144 exceed eps <= 134. All cases are excluded under the stated exact ownerless hypotheses.

G Complete local Z5 reduction at size84 (common source assumed)

The common integral source, complete targets and j = 5 are hypotheses. The goal of this file is to close all actual Z5 geometries, not merely scalar rows. Source notation is that of z5_alpha.md. Use

```
S=2K+1072-delta, A=I+tau+delta<=76,
K-delta>=220+I+tau, K-M>=R-74+A,
max(4,A-4)<=e_v<=241-A, L>=A+76,
delta>=10 implies L>=A+78.
```

Both leadings are active. The proof of S_v>M in the preceding tail report applies to every j5 source: r_v >= 25, R_o <= 118, e_o <= 241 give S_v-M >= 325 + 75-118 + 6-241 = 47. Hence empty integral ordinary centres are either LOW <= -M_v or HIGH >= S_v + M_o + 1. A retained translate with no core hit has at most four hits. After the eight O credits each retained singleton, and each licensed sole pair, has at least8 hits. Thus all of them have core hits and generic native endpoint errors <= xi_v (their clips are <= 68<72).

G.1 The base holes before outer integrality

The sharp raw caps are t_v <= 12 for active O_v and t_v <= 9 for active beta_v, H_v. Here t_v = d_v-M. Leading activity gives

```
t_v>=-7+I+tau+xi+2X_v+g_v.
```

An active O therefore forces tau <= 19 and 2X_v + g_v <= 19-I-tau-xi. If U_v or C6_o were active, q = X_v + g_v would be integral. The two opposite leading/core shortcuts, including the one-empty form, give

```
2q>=2(68-xi)-delta>=60+I+tau-2xi>=60+I-tau>=41.
```

This contradicts q <= 19. Similarly active V_v or C7_o makes q = X_v integral and gives the same contradiction. Thus active O_v forces the four distinct singleton holes U_v, V_v, C6_o, C7_o. Two active O’ s would force eight holes; so O_o is empty. These are all five holes, leaving V_o, C7_v retained. The corresponding two stars give X_o >= 21, whereas active C7_v gives t_v >= 2X_v + X_o + g_v + 1>22. This contradicts t_v <= 12. Both O’ s are empty.

If beta_v, H_v are active, the same argument with tau <= 16 forces four lower witness holes in addition to the two O holes. Thus they cannot both be active. No lower coordinate integrality was assumed in this argument: an active witness itself licenses the needed integer displacement.

G.2 Integral coordinates and the literal grammar

The unchanged four-coordinate identities in Appendix E of the retained report now prove outer integrality. The needed numerical rules are checked here rather than imported from size85. An empty cross above an active cross is HIGH and makes the leading lose at least r_v + 80, which exceeds delta <= 76. Thus P_v active = > C5_v = > C6_v = > C7_v. An empty integral own centre above a retained own centre is HIGH; any retained larger centre then loses72 points, more than epsilon <= 68. Thus missing V forces U, H missing; missing U forces H missing.

Pair1 has a sole actual owner whenever retained, because P_s + P_t = -(s2 + s3 + s4 + t2 + t3 + t4)<-3M whereas active centres exceed-M_v. The additional unchanged identities are

```
beta_s+beta_t+C5_s+C5_t=-2(s3+s4+t3+t4),
P_s+beta_s+C5_t=-2s3-2s4-t2-t3-t4.
```

They exclude the indicated simultaneous activities. In A0 the beta pair and both O' s are holes. A nonintegral v3 forces U_v, C6_o holes, and a nonintegral v4 forces V_v, C7_o holes. The remaining active H_v, alpha_v and other lower own centre contradict respectively v3 = H_v-V_v-alpha_v or v4 = H_v-U_v-alpha_v. Nonintegral v2 would cost H_v, alpha_v, C5_o. Thus all coordinates are integral. The chains retain V_s, V_t, C7_s, C7_t. If P is retained the thirteen exact A0 templates are those listed in the full-rank checker (eleven endpoint cards with the three orientations of A). If P is empty, the final hole, up to reflection, is H_s, alpha_s or C5_s.

For split beta, the identities force exactly O_s, O_t, H_s, H_t, C5_t as holes, with sole P owner s. The retained lower own centres and alphas prove integrality. The split-beta transport below excludes this geometry using only the whole beta capacity. For sole beta_s, base holes are O_s, H_s, O_t. A nonintegral lower coordinate again costs two extra holes. On t, H_t, alpha_t and the other lower own centre contradict it. On s, alpha_s, beta_s make d_s, s3 + s4, s2 integral and contradict it. Nonintegral t2 costs three holes H_t, alpha_t, C5_s. The chains retain the four extreme leaves. If P is empty, its final companion is precisely alpha_s, H_t, alpha_t, C5_s, C5_t or U_s. If P is retained, it is either opposite beta' s owner, the same owner with alpha_s retained, or the same owner with alpha_s, C5_t empty.

In B, the strict beta-sum sign forces C5' s pair retained with sole owner s. Any retained P has the same owner. The beta/H exclusions cost one hole on each sign in addition to the two O holes. The same two-witness identities and own/cross chains prove integrality and extreme-leaf retention. The remaining alternatives are alpha_s retained, or alpha_s the only extra hole; in the latter case P, alpha_t and all eight lower own/cross resources are retained. No empty pair receives a nominal owner.

G.3 Common endpoint and forest calculus

Write eps = delta-e. Every sole native resource has lower and upper bounds 72-r_v-l-xi_v and S_v + r_o-71 + u + xi_o. Distinct original resource deficiencies, or deficiencies of disjoint actual groups, sum to <= eps. Native endpoint weights are recorded explicitly below. For one actual Q of size q and overlap F, weighted stars and ordinary gap pairs give

```
cX_s+dX_t >=(c+d)F-max(c,d)eps/2,
cg_s+dg_t >=(c+d)F-max(c,d)eps/2.
```

These are separate budget consequences. Every group pair has at most one empty member; the full intersection in that case is paid solely by the retained partner' s off-target points. A retained own triple gives g + h >= q + F-Psi. It shares its actual deficiency with all of its subgroups. No subtree or subgroup is allocated twice.

G.4 A0

Retained P uses the thirteen actual templates and eleven exact affine cards A–K in `upper83_fullrank/ check.py`. That checker reconstructs the endpoint identities at d76 and proves margins 321, 141, 324, 59, 7, 167, 167, 25, 203, 257, 167. Their source, activity, core-hit and sole-ownership predicates have now been supplied above. For E the empty H_t is HIGH above retained U_t, providing credit11; the other cards use credit8. Card D uses both actual triples and one common support optimizing 5q + 6F.

When both pairs are empty and the extra hole is H_s or C5_s, the initial fixed ledger gives credit13 (HIGH11/LOW22/touched20 or HIGH8/LOW25/touched20). It needs only LOW paired centres and the leading inequalities; it does not require normal endpoints. The empty pairs are LOW because retained H or alpha lies above each beta, and a HIGH cross would lose r_v + 80 leading points. Put B = alpha_t, D = C5_t. The physical six-resource side gives S_t <= K +

616-e. From H_t = 2(a + b) + G + alpha_s + B + D, one generic lower endpoint and the actual B, D packing inequality give

```
M+2X+G+F <= R+484-e+u_Ht+l_alpha_s+2Gamma_BD+2xi.
```

Rebate those three disjoint groups in the stars and ordinary pairs. Their maximum coefficient is two, so M + 7F <= R + 484-e + 2eps + 2xi. Using L >= A + 78 and A <= 76 yields

```
2I+2tau-2xi+7F+3e<=482.
```

The full-rank kernel 2tau-2xi + 7F >= 444 makes the left side at least483. For the remaining alpha_s hole, `z5_alpha.md` proves the full ordered credit14->27->31->35->38->42 path and the final884>882 contradiction.

G.5 A1/B transports reconstructed at d76

The fresh physical inequalities are listed first. Every centre identity below follows from a = M_s + X_s, b = M_t + X_t and the definitions; the endpoint lists specify the actual resource accounting.

- Opposite owners beta_s, P_t: use V_t = 2d_s + beta_s + P_t + 2a + b + g_s, upper V_t, lower beta_s, P_t, and twice leading s. This gives S_s + 3g_s + 4X_s + 2a + b <= 924 + r_s + delta-2e + xi + xi_s.
- Same owners with alpha_s retained: the beta-sum identity and alpha_t>-b give alpha_s-P_s>beta_s + d_t + 2a + b + G. The upper alpha_s, lower P_s, beta_s and leading t give S_t + M_s + 2X_s + 3X_t + g_s + 2g_t <= R + 1012-r_t-e + 2xi_s + xi_t.
- Four endpoints: U_s = P_s + alpha_t + C6_t + 2a + 4b + 2G. The upper U_s and three lower endpoints give S_t + 2M_t + 2X_s + 4X_t + 2G <= 2R + 894 + 2r_t-e + xi_s + 3xi_t.
- HIGH alpha_s: use alpha_s + C5_t = d_t-2(a + b)-G, d_t = H_t-alpha_t, and alpha_s >= S_s + R_t + 1. The three retained endpoints C5_t, H_t, alpha_t give 2S_s + 2X + G <= 2R + 965-e + xi. Here e14 includes both O' s, HIGH H_s and HIGH alpha_s. The alpha HIGH guard follows from d_s-R_s >= 220 + I + tau-R_s>0 and the core hit of beta_s.
- Empty P, extra C5_s or U_s: sum upper alpha_s, H_t minus lower 2beta_s, alpha_t, C5_t. The literal expression is 2d_s + 2a + 2b + G. Twice leading s gives 2M + 6X_s + 2X_t + 3g_s + g_t <= 843 + delta-2e + 3xi.
- Empty P, extra H_t: upper alpha_s, V_t minus lower 2alpha_t, 2C5_s, 2beta_s, C5_t equals6a + 5b + 3g_s + 2g_t. Hence 4M_s + 3M_t + 6X_s + 5X_t + 3g_s + 2g_t <= 2R + 736 + r_s + delta-2e + 4xi + xi_s.
- Empty P, extra alpha_t: upper alpha_s, H_t minus lower 2beta_s, C5_t, 2C5_s equals6a + 3b + 3G + 3h_t. Hence 4M_s + M_t + 6X_s + 3X_t + 3G + 3h_t <= 2R + 746 + 3r_s + delta-2e + 5xi_s + 2xi_t. Use the H_t, U_t, V_t triple, two g_s pairs and separate stars, obtaining3q + 15F-6eps for the gap terms.
- Split beta: its entire hit count N has N >= 76-delta and beta_s + beta_t >= N-M. Use upper alpha_s, lower P_s and twice leading t in alpha_s-P_s = beta_s + beta_t + 2d_t + 2a + 2b + G. Conservatively, S_t + M + 2X_s + 6X_t + g_s + 3g_t <= 920 + 2delta + xi. Both H' s are HIGH, so credit14 is compatible. N is charged only to the one whole pair, never to either component separately.
- Branch B, alpha_s retained: alpha_s-C5_s>d_t + 2a + 2b + G. Two endpoints and leading t give S_t + M + 2X_s + 4X_t + g_s + 2g_t<R + 1017-e + xi.

Replace each S_v by T_(r_v) + tau_v + 4R_v + 6 + e_v. The exact checker uses the full source peel polygon and all-rank inverse mass profiles; the minima of left minus right are respectively

811/2, 203/2, 258, 230, 91/2, 1105/2, 305, 415, 223. The two endpoint sides retain physical names in these calculations; the scalar quadratic index ranges over both orientations.

These nine exclusions route all non-tail A1/B alternatives. HIGH-alpha handles extra alpha_s; the next three rows handle C5_s/U_s, H_t, alpha_t. The exact final empty-P C5_t family is already fully proved and pinned by `reports/upper83_tail_20261019.tex` and `verification/upper83_tail.py`. With P retained, use opposite owners, same owners with alpha_s retained, or the four-endpoint row if alpha_s is empty. Branch B uses its own row when alpha_s is retained and the four-endpoint row otherwise. Together with A0 this closes every Z5 geometry under the stated common source hypotheses. The common-source and positive-hole obligations are discharged separately in structure.md and positive.md.

H Size-84 Z5 alpha: order before normality

Local theorem, assuming the common integral source and the exact Branch-A whole holes {P, Bpair, O_s, O_t, alpha_s}. This file does not assert the global classification. All thirteen other singleton resources are retained. Notation follows the full-rank report: d = 76, m = 68, r_s + r_t = 67, I = (r_s-33)(r_s-34), A = I + tau + delta <= 76, R = 67 + xi, M_v = R_v + e_v, a = M_s + X_s, b = M_t + X_t, G = g_s + g_t. Write B = alpha_t, C = C5_s, D = C5_t. In particular B is a centre here, not the empty beta pair or a source fringe count.

H.1 The initial tail credit does not need normality

The generic Z5 core-hit argument gives each retained centre Z at least one core hit. The native rank endpoints are Z >= 72-r_v-l_Z-xi_v and Z <= S_v + r_o-71 + u_Z + xi_o whenever the selected native clip is below72. Also Z >= 1-R_v is always available from a core hit. The empty alpha_s and both empty beta components are LOW. The original initial label inventory therefore gives credit14.

With epsilon = delta-14, the full-rank stars and ordinary pairs give X_v, g_v >= F-epsilon/2. Since tau + 2F >= 134 (min(T_p-2p) = -1 on each side), X_v, g_v >= 36 + I/2, hence at least36. The retained {H_s, U_s} pair gives h_s >= F-epsilon >= 6 + I, using tau + F >= 68. LOW alpha_s gives h_s <= X_s. K-delta >= 75 + I + tau gives d_v >= 75 + I + tau + g_v + 2X_v.

These inequalities license the complete32-type tail inventory from upper84/ audit/ verification_program/ proof/ report.tex (the paragraph “The alpha_s hole: the fixed-credit argument”). Here are its size-specific strict checks. On s, six beta types lie below d_s-X_s. The remaining 26 types give S_s <= d_s + 26. The three labels C16_s, C17_s, C18_s exceed that cap since C + s3 + h_s >= 1 + e_s + X_s + g_s + h_s >= 83>26. At d_s + 1, d_s + 2 every remaining type except the sole score O_s-t1 is either <= d_s or >= d_s + 6; use h_s >= 6, g_s >= 36 and d_s>g_s + 2h_s. Thus S_s <= d_s + 1. Above H_s + M_s the only possible types are d_s-g_s, d_s, O_s-t1; they cannot cover three consecutive integers. Therefore S_s <= H_s + M_s + 2. On t, sixteen types lie below H_t + M_t and all nonfixed resources end there, so S_t <= H_t + M_t + 16. X_t >= 36 makes C22_t, C23_t, C24_t HIGH; D + M_t + 2X_t + g_t >= 1 + e_t + 2X_t + g_t >= 113>16 makes C16_t, C17_t, C18_t HIGH. The s side gives the analogous six labels and C33_s. Thus HIGH21, LOW29, touched23 gives credit27, before normality.

H.2 Separate the two order tests

At credit e >= 27, epsilon = delta-e <= 49. All clips are below72, so generic errors <= xi_v apply even to centres outside [1, S_v]. The source lemma gives L >= A + 78 since delta >= 27. Grouping whole actual targets gives S_s <= K + 540-e and S_t <= K + 616-e. Consequently the exact identities H_s = a + 2b + G + h_s + B + C and H_t = a + 2b + G + h_s + B + D imply

```
M_t+X_s+2X_t+G+h_s+B+C <= R+490+r_t-e+u_Hs+xi_t,
M_t+X_s+2X_t+G+h_s+B+D <= R+566+r_s-e+u_Ht+xi_s.
```

Choose ONE actual truncated core maximizing q + 6F. The weighted inverse staircase gives, throughout tau <= 49,

```
q+6F+3tau-xi >=466,
q+6F+2tau-xi >=458.
```

The checker reconstructs these minima using integer mass constraints. The actual triple {H_s, U_s, V_s} and the two g_t pairs give G + h_s >= q + 2F-epsilon; stars give2X >= 4F-epsilon.

If B >= a, lower C generically in the FIRST physical row. As r_s + r_t = 67,

```
M+2X+G+h_s <= R+485-e+epsilon+xi.
```

Combining the preceding bounds, L >= A + 78 and delta <= 76-I-tau gives

```
q+6F+3I+3tau-xi+4e <=559.
```

At e = 27 its left side is at least466 + 108 = 574>559. Hence B<a. This conclusion uses the smaller physical-s capacity; replacing both physical rows by the larger one loses the required margin.

The six new LOW labels U5_t, U6_t, U7_t, U8_t, V3_t, V4_t meet two new groups. For V3_t, V4_t use D >= 1-R_t and b>R_t; positivity of D is unnecessary. Credit is now31.

Independently C<b. If C >= b, it is the smallest retained ordinary s centre and the leading starts at C + 2X_t + B + g_t>C. For the strict sign, already at credit27, 2X_t + g_t >= 127.5 whereas R_t <= R <= 116, and B >= 1-R_t. Thus [1, C-M_t-1] is fixed-only. The same credit27 stars give X_t >= 42.5, so it contains at least42 integers. But the s-side fixed inventory has11 HIGH and18 LOW among55 labels, leaving at most26. Contradiction. C<b gives C66_s, C67_s, C68_s, C35_s LOW; for C35_s = C-D-2b-g_t-h_s use D >= 1-R_t. No new group is touched. Credit is35.

Finally suppose D >= a. Lower B generically in the SECOND physical row:

```
M+2X+G+h_s <= R+561-e+u_Ht+l_B+xi.
```

Let N = Gamma_Ht + Gamma_B. The triple, g_t pairs and stars used above avoid both resources, so their respective budgets may each be epsilon-N. As u_Ht + l_B <= N, this gives

```
L+q+6F <=561-e+2epsilon-N+xi <=561-e+2epsilon+xi,
q+6F+2I+2tau-xi+3e <=559.
```

At e35 the latter left side is at least458 + 105 = 563>559. Hence D<a. The three labels C66_t, C67_t, C68_t add credit3, giving38. C25_s, C25_t, C35_t, C45_s, C45_t add credit4, giving42. Their signs use C<b, D= 1-R_s, D >= 1-R_t, not positivity. The two C25 labels touch the sole remaining baseline group.

H.3 Normality and terminal

Only now exclude BAD centres. Put r_max = 34 + k, I = k(k + 1). A BAD retained singleton loses at least72-r_max = 38-k, whereas epsilon <= 34-I-tau. Their difference is k^2 + 4 + tau>0. Thus all retained singleton centres are normal.

The exact transport of the full-rank report now applies with the actual source fringe count B_fr = B_s + B_t:

```
2M_t+2q+11F <=2R+778+2B_fr+2r_t-delta+5(delta-42).
```

For completeness its eleven distinct singleton groups are {B, D}, {H_s, U_s, V_s}, {U_t, V_t}, {C6_s, C7_s}, {C}, {H_t}; the separately valid weighted stars use the same actual core. The native losses are not counted as disjoint between these two separate inequalities. At credit42 all required gaps first satisfy >= tau. The boundary-safe complete-core intersection raises them to >tau. The complete-core long-gap intersection and three-translate union lemma therefore permits q = F = 68 simultaneously in those groups. Use e_t >= Theta-d-12 and Theta = 84 + A, then retain the actual fringe in K <= M + R + 1 + B_fr. The transport becomes

```
4I+4tau-2xi_s+13*68 <=542+2(Theta+B_fr)<=882.
```

The left side is at least884, contradiction. Thus this exact local alpha-hole family is excluded, without an early normality assumption.